Vataleni PMSG SSO Model Guide

How to run, inspect and interpret the self-contained Simulink model. This guide describes the saved 21 October 2026 revision.

Model: **model/Vataleni_PMSG_SSO_Final.slx** | MATLAB/Simulink R2025b (25.2) | Solver: fixed-step ode4, 0.0002 s, 12 s.

Open and run

Open the SLX in MATLAB R2025b with Simulink installed, then press **Run**. No project script, base-workspace variable, path addition, or external data file is required. The default run is the strong-grid baseline. The 35-channel simulation log is written to **simlog** in the SimulationOutput object when launched programmatically.

Choose a study case

In Model Explorer, select the model's **Model Workspace**. Edit the three variables below, save the model, then press Run. The SCR step occurs at 2 s. Only the weak-grid cases excite the benchmark SSO mode.

Case	scr_initial	scr_final	mitigation_enable
Baseline	10	10	0
Weak-grid SSO	10	5	0
Mitigated weak grid	10	5	1

Subsystems and signals

Subsystem	Role
Wind Turbine + PMSG	Aerodynamic Cp law, rotor inertia and torque balance
MSC Control	Speed PI and averaged q-axis current response
DC Link	Capacitor energy balance
GSC Control	DC-voltage and d/q current PI loops
PLL	PCC-voltage phase tracking and selectable integral gain
Grid Current Dynamics	Averaged d/q filter and Thevenin grid
PCC Interface	PCC voltage and active/reactive power
Measurements and Analysis Interface	Diagnostic vector, abc reconstruction and Scopes

Engineering Scopes

Double-click **Measurements and Analysis Interface**, then **Engineering Scopes**. The Scope names identify their units. Multiple quantities share an axis only when they use the same per-unit base.

Scope group	Signals
Wind and rotor	Wind speed; rotor speed; mechanical and electromagnetic torque
Machine-side control	q-axis current and current reference
DC link	DC voltage
Grid-side control	d/q current, references and converter commands
PCC and power	PCC d/q voltage and magnitude; active and reactive power
PLL and SSO	Estimated frequency, phase error and active-power oscillation

Model equations and fidelity

The governing equations and parameter provenance are listed in MODEL_ARCHITECTURE.md, docs/MODEL_EQUATIONS.md and docs/PARAMETERS.md. Seven physical/control equation groups are connected native Sum, Gain, Product, Divide, Saturation and Trigonometric blocks with native Integrators for the eleven states. One embedded MATLAB Function block assembles the diagnostic vector and reconstructed phase waveforms.

The machine-side converter is a **reduced-order q-axis torque-current equivalent**. This model does not resolve stator d/q voltage dynamics, switching devices, PWM ripple, machine saliency or shaft torsion. Results are a benchmark of converter-grid interaction under the documented assumptions; they are not a validation of a specific physical wind farm.

Optional parameter reference

parameters/VaxSim_Parameters.m contains an editable parameter list and three scenario presets. The SLX does not call this file. Editing the file alone does not change a simulation; make parameter changes in the corresponding model blocks or Model Workspace variables and save the SLX.

Dependencies

The saved model was executed with MATLAB and Simulink R2025b (25.2). It uses standard Simulink blocks and one internal MATLAB Function block, which uses the installed Stateflow engine. Simscape Electrical is installed but is not used by this reduced-order model. No external custom S-functions or referenced models are required.